HAZEL NOVERO WINATA



DATA SCIENCE | DATA ANALYST | DATA ENGINEER | BUSINESS ANALYTICS

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PROFESSIONAL PROFILE

BINUS University Data Science undergraduate (GPA 3.53) focused on Data Science, Data Analysis, Data Engineering and Business Analytics. Combines predictive modeling, large-scale healthcare and financial-data analytics, cloud ML deployment, and database/system design, with documented work spanning 1.21M healthcare records and 319,978 vehicle-financing transactions. Adds nearly two years of leadership, public speaking and cross-functional communication through BINUS University Learning Community outreach across cities outside Greater Jakarta.

EDUCATION

BINUS University | Bachelor of Data Science (in progress) **GPA 3.53 / 4.00**

Relevant coursework: Machine Learning, Deep Learning, Model Deployment, Database Technology, Data Management & Organization, Big Data & Infrastructure, Program Design Method.

EXPERIENCE & LEADERSHIP

BINUS University Learning Community (BULC) | Promotion Representative & Team Lead (Part-Time) **Nearly 2 years**

- Chosen as the sole student representative assigned alongside BINUS marketing staff to priority regional outreach locations, including Pekanbaru, Riau; represented the university in school and prospective-student engagements.
- Presented admissions, study programs and scholarships at approximately 30 schools across several provinces outside Greater Jakarta; adapted public-facing messages to students, parents and school stakeholders.
- Led trip preparations, student-team coordination and event delivery, collaborating with BINUS marketing/finance personnel and school partners on logistics and follow-up.

Data Science Club, BINUS University | Event Organizing Committee (Active Member) **Approx. 1 year**

- Supported the organization of technical seminars, student community events and company visits, assisting with event logistics, participant coordination and communication.

Kindergarten Teaching Facilitator | Classroom Teaching Assignment **May–Jun 2025**

- Delivered five early-childhood learning sessions, coordinating classroom activities and explaining concepts in age-appropriate language.

SELECTED PROJECTS

Patient Risk Journey Clustering | Python, Pandas, Data Integration, Scikit-learn, StandardScaler, K-Means

- Integrated multi-table healthcare data and analyzed 1.21M filtered encounter records; engineered 22 patient-level features for 57,386 profiles, evaluated clustering candidates and interpreted five exploratory healthcare journey segments.

Credit Score Prediction & AWS ML Deployment | Grade A - Model Deployment | Python, XGBoost, LightGBM, Scikit-learn, MLflow, AWS SageMaker, EC2, Streamlit

- Compared eight classifier/feature-set configurations with customer-disjoint evaluation (best recorded macro F1 0.6898); separately built an OOP training/evaluation pipeline, tracked experiments in MLflow and demonstrated AWS-hosted model training and inference.

Customer Repeat-Order & Transaction Quality Analysis | Python, Pandas, NumPy, EDA, Data Validation, Customer Analytics

- Audited 319,978 automotive financing transactions and identified 2,341 exact duplicates; investigated customer identifiers, demographic anomalies, price/tenor checks, geographic category inconsistency and repeat-order definitions.

InkluMap - Inclusive Learning Support Analytics | Python, Gower Distance, Hierarchical Clustering, FastAPI, PostgreSQL, Next.js

- Developed an accessibility-focused decision-support prototype from 120 synthetic student records, combining needs-based segmentation, interpretable support recommendations and a database-backed dashboard. Passed campus selection and advanced to the national competition stage.

Overhead-MNIST Autoencoder & Image Reduction | TensorFlow, Keras, CNN, Autoencoder, SSIM, Validation Tuning

- Trained 128-dimensional image-reconstruction models on 16,016 overhead images; evaluated baseline and nine modified configurations, raising held-out SSIM from 0.4864 to 0.7573 (+55.7% relative) with qualitative reconstruction review.

Spaceship Titanic MLflow & AWS Cloud Workflow | Grade A - Model Deployment | Python, XGBoost, LightGBM, Optuna, MLflow Registry, SageMaker, EC2

- Engineered passenger features and optimized supervised classifiers (tuned XGBoost five-fold CV accuracy 0.81261); documented nested MLflow experiments, champion model selection, a SageMaker endpoint and EC2-hosted Streamlit inference.

Student Placement & Salary ML Application | 100/100 - Course Midterm | Python, Scikit-learn, LightGBM, MLflow, FastAPI, Pydantic, Streamlit

- Built classification and salary-regression pipelines from 5,000 student records; tracked 26 MLflow runs and implemented an API-based inference interface. Documented that the placement target follows a deterministic rule in this academic dataset.

Weather-Traffic Big Data & API Streaming | Python, PySpark, Structured Streaming, REST APIs, JSON, OpenWeather, TomTom

- Collected timestamped weather and traffic feeds at sampled road coordinates in 10 Indonesian cities; ingested API JSON snapshots using Spark Structured Streaming, joined sources by location/time and audited 740 joined rows for possible many-to-many duplication.

BNPL Relational Database & ERD | SQL, MariaDB, 1NF-3NF, Functional Dependencies, PK/FK, ERD

- Designed and normalized a synthetic Buy Now Pay Later database into six relational tables covering 50 purchases and 333 installments; modeled entity cardinalities, insertion/update/delete anomalies and gaps between conceptual and enforced foreign keys.

UML & Business Process/System Modeling | UML Use Case, Class, Sequence Diagrams, ERD, REA, Visio, draw.io

- Modeled tutoring and business-operation workflows using use-case and sequence diagrams, class relationships (association, aggregation, composition, realization), actors, requirements, database ERDs and REA-oriented business processes.

TECHNICAL & PROFESSIONAL SKILLS

Analytics & ML: Python, SQL, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, feature engineering, EDA, classification, regression, K-Means, model validation.

Big Data & Cloud: PySpark, Spark Structured Streaming, REST/JSON ingestion, API polling, data cleaning/integration; AWS SageMaker training/endpoints, EC2 hosting, cloud inference.

Deep Learning & Deployment: TensorFlow/Keras, CNN, LSTM, Autoencoder, conditional GAN; Optuna, MLflow tracking/registry, FastAPI, Pydantic, Streamlit, Git/GitHub.

Database & Business Modeling: PostgreSQL, MariaDB, SQL joins, normalization 1NF-3NF, primary/foreign keys, ERD, UML use-case/class/sequence diagrams, REA, process and requirements modeling.

Professional: Leadership, stakeholder communication, public speaking, presentation delivery, educational outreach, team coordination.

CERTIFICATIONS & RECOGNITION

NVIDIA Deep Learning Institute | Fundamentals of Deep Learning - Certificate of Competency (Mar 2026)

Competition recognition | Finalist, SPARC DataFest BINUS 2026; finalist, ASA DataFest BINUS 2026; InkluMap selected to advance from campus to national stage.